

IST 606: Security Management and Computer Forensics

Detailed Essay Answers — All Four Practice Sets

This document provides **complete, fully written essay answers** (not outlines) for all twenty essay questions across the four practice sets. Each answer is structured to demonstrate the depth, argumentation, and use of evidence expected at the marks level indicated.

SET 1 — Crimes Against Persons

Essay 1 (17 marks)

Discuss the concept of cybercrime against persons. Using SIM card registration fraud as a case study, explain how weaknesses in identity verification processes create opportunities for cybercriminals, and propose THREE measures that telecom operators could implement to close these gaps.

Cybercrime against persons refers to the category of digital offences in which the primary target and victim of harm is an individual, rather than an organisation, government body, or society at large. Under Cameroon's Law No. 2010/012 of 21 December 2010 on Cybersecurity and Cybercriminality, such offences include identity theft, online harassment, cyberstalking, defamation, mobile money fraud, and unauthorized access to personal accounts. What unites these offences is that the harm — financial loss, psychological distress, reputational damage, or loss of personal safety — is experienced directly and personally by an identifiable victim. This distinguishes crimes against persons from crimes against property (which target organisational or financial assets), crimes against society (which threaten collective institutions such as elections), and crimes against government (which target state infrastructure).

SIM card registration fraud is one of the clearest and most consequential examples of cybercrime against persons in the Cameroonian context, precisely because mobile phone numbers have become the backbone of personal financial identity through mobile money platforms such as MTN Mobile Money and Orange Money. SIM registration was introduced as a security measure: by requiring every SIM card to be linked to a verified national identity document, regulators intended to make telecom usage traceable and accountable, thereby deterring anonymous criminal use of mobile networks. In principle, this ties every phone number to a real, verifiable person.

In practice, however, the verification process at the point of registration is frequently weak, and it is precisely this weakness that cybercriminals exploit. Several specific vulnerabilities recur. First, registration agents — often operating from small kiosks with high customer volume and minimal supervision — do not always rigorously compare the photograph on a submitted ID card against the physical appearance of the person presenting it, nor do they verify that the ID itself is authentic rather than a photocopy or forgery. Second, once an ID copy has been submitted for one transaction, there is frequently no policy requiring its secure destruction or restricted storage; copies may be kept loosely, photographed, or even resold, allowing the same identity document to be reused fraudulently for an entirely unrelated and unauthorized SIM registration months later. Third, there is generally no secondary or biometric confirmation step — such as a verification call, SMS code

sent to the legitimate ID holder's existing registered number, or biometric fingerprint matching — that would catch a fraudulent registration before it is finalised and activated. Together, these gaps mean that the security premise of SIM registration (that a SIM is reliably linked to its true owner) can be circumvented with nothing more sophisticated than a photocopied document and a cooperative or inattentive agent.

Once a fraudulent SIM has been registered in a victim's name, the consequences cascade quickly. The criminal can use the SIM to receive SMS-based one-time passwords, enabling them to take control of the victim's mobile money account, banking applications, or social media profiles that rely on phone-based two-factor authentication. Because the SIM is registered to the victim's real identity, suspicion is initially deflected away from the true perpetrator, and the victim may not discover the fraud until financial loss has already occurred.

To close these gaps, telecom operators could implement three concrete measures. First, **mandatory biometric verification at the point of registration**, such as fingerprint or facial recognition matched in real time against the national ID database, would make it substantially harder to register a SIM using someone else's documents alone, since the physical biometric of the person present would need to match the identity being claimed. Second, **real-time cross-checking against a centralized national identity database**, so that any new SIM registration automatically flags if the same ID number has already been used for an active SIM, would prevent the silent re-use of a previously submitted document for a second, unauthorized registration. Third, **mandatory SMS or call-based confirmation to any existing registered number associated with that identity** before a new SIM tied to the same ID is activated, would give the legitimate ID holder a final opportunity to detect and block fraudulent registration before it succeeds. Implemented together, these three measures would shift SIM registration from a paperwork-based system vulnerable to human inattention toward a layered, verifiable process consistent with the security objectives the registration requirement was designed to achieve.

Essay 2 (17 marks)

With reference to a real or hypothetical spear phishing attack on an organisation's email system, explain how such an attack is typically structured, the human and technical vulnerabilities it exploits, and the digital forensic tools that would be used to investigate it.

Spear phishing is a targeted form of phishing in which an attacker crafts a deceptive email aimed at a specific individual or small group within an organisation, rather than broadcasting a generic message to a mass audience. Unlike conventional phishing, which relies on volume and the statistical likelihood that some recipients will fall for a generic lure, spear phishing relies on personalisation and credibility: the attacker researches the target, often using publicly available information from LinkedIn, company websites, or social media, to craft a message that appears to come from a trusted colleague, supplier, or senior executive.

Consider a hypothetical case in which the finance department of a Cameroonian company receives an email that appears to come from the company's Chief Executive Officer, requesting an urgent wire transfer to a "new supplier account" ahead of a board meeting. The email uses the CEO's actual name and writing style, references a real ongoing project, and is sent from a domain that differs from the company's real domain by only a single character (a classic case of domain spoofing or typo-squatting, similar in principle to the website-cloning techniques used against the Cameroon passport portal). The finance officer, under time pressure and accustomed to receiving instructions from senior management, processes the transfer without verifying it through a secondary channel.

This attack structure exploits both human and technical vulnerabilities simultaneously. On the human side, it exploits the psychological principles of authority (compliance with an apparent instruction from a superior), urgency (the pressure of an impending deadline discourages careful verification), and routine trust (employees are conditioned to act on emails that look legitimate and arrive through the normal communication channel). On the technical side, it exploits weaknesses such as the absence of strict email authentication protocols (SPF, DKIM, and DMARC, which would normally help flag a spoofed sending domain), the lack of multi-factor authentication or out-of-band verification requirements for financial transactions, and insufficiently configured spam/phishing filters that fail to catch a near-identical lookalike domain.

Investigating such an attack draws on several digital forensic tools, each suited to a different stage of the inquiry. **MxToolbox** would be used early in the investigation to inspect the email headers and verify the sending domain's authentication records, quickly revealing whether SPF/DKIM checks failed and confirming the spoofing. **Wireshark** could be used to capture and analyse any network traffic generated if the phishing email contained a malicious link or attachment that "phoned home" to an external server, helping to identify the command-and-control infrastructure behind the attack. **Gophish**, while primarily a phishing simulation tool, would be valuable in the aftermath of the incident to test staff susceptibility and measure the effectiveness of any new training introduced as a corrective measure. **Splunk** would be used to search and correlate logs across the email server and financial systems, identifying the precise time the malicious email was received, opened, and acted upon, and cross-referencing this with any unusual login or transaction activity in the same window. Finally, **Autopsy** would be used to examine the finance officer's computer for any malicious attachments, browser history showing the spoofed domain being visited, or evidence of credentials being entered into a fraudulent login page, while careful imaging and hashing of the device would ensure any findings remain admissible if the case proceeds to prosecution.

Taken together, this case illustrates that spear phishing is rarely defeated by technology alone; because it is engineered to exploit predictable human responses to authority and urgency, effective defence requires a combination of technical controls (email authentication, multi-factor verification for sensitive transactions) and organisational policy (mandatory secondary verification for financial instructions, regular staff awareness training) — with digital forensics providing the after-the-fact reconstruction needed both to understand what happened and to support any legal action against the perpetrators.

Essay 3 (16 marks)

"The success of most cybercrimes against persons depends more on human error than on technical sophistication." Critically discuss this statement with reference to at least TWO categories of cybercrime against persons (e.g. SIM swap fraud, mobile money scams, identity theft).

This statement captures an important and well-supported truth about cybercrime against persons, though it requires some nuance to avoid overstating the case. The available evidence from both the SIM card registration fraud case and broader patterns in mobile money scams strongly supports the view that human and procedural failures, rather than advanced technical exploitation, are the primary enablers of these crimes — but it would be inaccurate to suggest that technical capability plays no role at all.

Consider first SIM swap fraud, as examined in the case of Marie's stolen identity. The attacker in that scenario did not need to breach any encrypted system, write any malicious code, or exploit any software vulnerability. Instead, the entire crime depended on a single human failure: a registration agent's insufficient diligence in verifying that the person presenting a photocopied ID card six months earlier was the same person now

requesting a new SIM in that name. No firewall was breached; no password was cracked. The "vulnerability" exploited was institutional carelessness in document handling and identity verification — a profoundly human and procedural weakness rather than a technical one. Similarly, in many mobile money scams across Cameroon, fraudsters succeed not through hacking mobile money platforms' backend systems (which are typically well-secured by the telecom operators) but through social engineering: calling victims, posing as customer service agents, and persuading them to read out a one-time password or PIN under the pretext of resolving an account issue. Again, this is a human vulnerability — trust, lack of awareness, susceptibility to authority — rather than a cryptographic or system-level flaw.

This pattern is consistent with broader cybersecurity research showing that social engineering remains the dominant vector for cybercrime against persons precisely because human psychology is a more reliable and lower-cost target than technical defences, which have grown increasingly sophisticated and difficult to penetrate directly. It is generally far easier to deceive a person into voluntarily disclosing information or approving a transaction than to defeat encryption or bypass authentication systems through pure technical means.

That said, the statement should not be read as implying that technical sophistication is irrelevant. Some forms of cybercrime against persons do require genuine technical capability: more advanced SIM swap attacks, for instance, can involve exploiting weaknesses in telecom signalling protocols (such as SS7 vulnerabilities) to intercept calls and messages without any direct social engineering of an agent at all. Identity theft increasingly involves the use of sophisticated phishing kits, malware, or even AI-generated deepfake voice cloning to impersonate victims convincingly. In these cases, while human error (such as a victim clicking a malicious link) often still provides the entry point, the subsequent exploitation may rely on real technical skill in areas like cryptography, network protocol manipulation, or malware development.

The most defensible synthesis, therefore, is that human error — whether on the part of the victim, an intermediary such as a registration agent or telecom employee, or an organisation with weak verification procedures — is overwhelmingly the *initial enabling factor* in the large majority of cybercrimes against persons, including SIM swap fraud and mobile money scams as examined here. Technical sophistication, where it appears, typically serves to *scale up or refine* an attack that still ultimately depends on some human or procedural weak point being present in the first place. This has direct practical implications: investment in staff training, identity verification procedures, and public awareness campaigns is likely to yield greater protective value against crimes against persons than purely technical security investment alone — though both remain necessary as complementary layers of defence.

Essay 4 (17 marks)

Explain the digital forensic investigation process from identification through to presentation of findings. Using an identity theft scenario of your choice, illustrate how each stage would be carried out, and discuss the chain-of-custody requirements that must be observed throughout.

The digital forensic investigation process is typically described through a structured sequence of phases — identification, preservation, collection, examination, analysis, and presentation — each of which serves a distinct purpose and must be carried out rigorously to ensure that the resulting evidence is both technically sound and legally admissible. To illustrate this process concretely, consider the scenario of Marie, whose national identity card was used without her knowledge to fraudulently register a SIM card later used to drain her mobile money account.

Identification is the first phase, in which investigators determine that an incident has occurred and establish its scope. In Marie's case, this begins the moment she reports the unauthorized withdrawal to the police and ANTIC. Investigators must identify which systems and records are relevant: the telecom operator's SIM registration database, the mobile money platform's transaction logs, and any physical devices (the fraudulent SIM's host phone, if recoverable) that may hold evidence. At this stage, investigators also identify the legal basis for proceeding — confirming that Marie's complaint constitutes a credible allegation of cybercrime under Cameroonian law, justifying further investigative steps.

Preservation follows immediately, and its purpose is to ensure that no evidence is lost, altered, or destroyed before it can be properly collected. In this scenario, this would mean immediately requesting that the telecom operator preserve (rather than overwrite or purge) the relevant registration and transaction logs, and, if a suspect device is identified, ensuring it is seized and placed in a state (such as airplane mode or a Faraday bag) that prevents remote wiping or further data alteration. Preservation is critical because digital evidence is inherently fragile — logs can be automatically deleted on a rolling basis, and a powered device can be remotely wiped if not properly isolated.

Collection is the active gathering of the preserved evidence using forensically sound methods. This would involve formally requesting and obtaining certified copies of the SIM registration record (including the submitted ID photocopy) and the mobile money transaction logs from the telecom operator, and, for the suspect's device, using a tool such as Cellebrite UFED to perform a full physical extraction. Collection must be done in a way that does not alter the original data — for example, using write-blockers when copying any storage media, and working from forensic images rather than the original device wherever possible.

Examination involves the technical processing of the collected data to surface relevant material. For the extracted phone data, this might involve searching for any messages or call records connecting the device to the registration agent or to coordination around the mobile money withdrawal, including recovering any deleted messages using the device extraction tool's recovery capabilities. For the transaction logs, examination would involve isolating the specific withdrawal transaction, the agent ID involved, and any linked account activity in the surrounding time window.

Analysis is the interpretive phase in which investigators draw conclusions from the examined data. Here, the investigator would correlate the timestamp of the fraudulent SIM registration with the timestamp of the original ID photocopy submission six months earlier, establishing the link between the earlier transaction and the present fraud. The analysis would also assess whether the pattern of behaviour (timing, location, method) suggests an individual opportunist or part of a wider organised fraud ring operating through the same registration agent.

Presentation is the final phase, in which findings are compiled into a clear, structured report suitable for use by prosecutors and, ultimately, a court. This report must explain, in accessible language, what evidence was found, how it was obtained, and what conclusions it supports, while being precise enough to withstand cross-examination on technical grounds.

Throughout all six phases, **chain-of-custody requirements** must be rigorously observed. Chain of custody refers to the unbroken, documented record of who collected each piece of evidence, when, how it was stored, who had access to it at every point, and any transfers of custody between individuals or institutions. In Marie's case, this would mean documenting precisely when the telecom operator's logs were obtained and by whom, recording the exact time the suspect's device was seized and by which officer, logging every instance the device or any forensic image of it was accessed or analysed, and ensuring that the device itself is stored in a secure, access-controlled evidence locker between examinations. Any gap or inconsistency in this chain — for instance,

an unexplained period where the device was accessible to unauthorized personnel — creates an opportunity for defence counsel to challenge the integrity of the evidence and potentially have it excluded, regardless of how compelling the underlying findings might be. Chain of custody is therefore not a bureaucratic formality but the legal backbone that connects rigorous technical work to a successful prosecution.

Essay 5 (16 marks)

Discuss the skills and prerequisites required for a person to work effectively as a digital forensic investigator handling crimes against persons. Your answer should address technical skills, legal knowledge, and ethical considerations.

Working effectively as a digital forensic investigator, particularly in cases involving crimes against persons such as identity theft, SIM fraud, or mobile money scams, demands a combination of technical competence, legal literacy, and a strong ethical foundation. No single one of these dimensions is sufficient on its own; a technically brilliant investigator who is ignorant of legal procedure can produce evidence that is accurate but inadmissible, while a legally astute investigator without technical skill cannot properly extract or interpret the evidence in the first place.

On the **technical side**, an investigator must possess a solid understanding of how digital systems work at multiple levels: operating systems (file systems, registries, log structures), mobile device architecture (SIM data, app data storage, deleted-data recovery), and networking fundamentals (how data moves across telecom and internet infrastructure, and how this can be captured and analysed). Practical proficiency with forensic tools is equally essential — for crimes against persons specifically, this includes mobile extraction tools such as Cellebrite UFED (for recovering data from seized phones), imaging tools such as FTK Imager (for creating verified, unaltered copies of storage media), and analysis platforms such as Autopsy (for examining file systems and recovering deleted content). Beyond tool operation, the investigator needs genuine analytical skill: the ability to look at a transaction log, a registration record, and a device extraction, and recognise patterns, inconsistencies, or correlations that point toward what actually happened — for instance, noticing that a SIM registration timestamp aligns suspiciously closely with an earlier, unrelated transaction involving the same identity document.

On the **legal side**, an investigator handling crimes against persons must understand Cameroon's Law No. 2010/012 on Cybersecurity and Cybercriminality, including what specifically constitutes an offence, what evidentiary standards apply, and what procedural safeguards (such as obtaining proper authorisation before accessing private records) must be observed. Legal knowledge also encompasses an understanding of chain-of-custody requirements, the rules of evidence in Cameroonian courts, and the practical realities of jurisdiction — recognising, for example, that if a suspect or relevant data is located outside Cameroon, additional legal mechanisms (such as mutual legal assistance requests) will be needed, and that evidence collection methods must be defensible under cross-examination rather than merely technically correct.

On the **ethical side**, the stakes in crimes against persons are particularly sensitive because the investigation often involves highly personal information — financial records, private communications, identity documents, and sometimes deeply distressing content if the crime involves harassment or exploitation. An effective investigator must maintain strict confidentiality, ensuring that sensitive personal data uncovered during the investigation is not disclosed beyond what is necessary for the case. They must also maintain objectivity, presenting all relevant findings — including those that might complicate the prosecution's narrative — rather than selectively reporting only supportive evidence; integrity of this kind is what gives forensic findings their

credibility in court. Investigators must additionally be sensitive to the victim's situation: in a case like Marie's, where financial loss may be intertwined with significant personal stress, the investigator's professionalism and discretion can materially affect the victim's experience of an already difficult process. Finally, ethical practice requires investigators to recognise the limits of their own expertise and jurisdiction, escalating or seeking specialist support (for example, from ANTIC or international partners) rather than overstepping their competence or authority in pursuit of a result.

Taken together, these three dimensions — technical capability, legal literacy, and ethical discipline — form an interdependent foundation. A digital forensic investigator handling crimes against persons must be simultaneously a skilled technician, a careful student of the law, and a trustworthy custodian of other people's most sensitive information.

SET 2 — Crimes Against Society

Essay 1 (17 marks)

Discuss the concept of cybercrime against society, distinguishing it clearly from cybercrime against persons and cybercrime against property. Use election interference as your primary illustrative example.

Cybercrime against society refers to digital offences whose principal harm is directed not at a single identifiable individual, nor at a specific organisation's assets, but at the collective institutions, trust structures, and stability that hold a society together — most prominently, democratic processes, public order, and shared social cohesion. This category sits conceptually apart from the other two main classifications typically studied alongside it: cybercrime against persons, where the victim is an individual whose identity, privacy, or personal finances are directly harmed (as in SIM card fraud or identity theft), and cybercrime against property, where the target is a tangible or intangible asset such as money, intellectual property, or organisational data (as in website cloning or database theft).

Election interference is perhaps the clearest and most consequential illustration of cybercrime against society. Consider a scenario in which, following a presidential election in Cameroon, several polling stations report turnout figures exceeding the number of registered voters, while simultaneously a wave of AI-generated deepfake videos depicting a candidate making inflammatory remarks spreads virally across social media, triggering public protests in multiple regions. No single victim can be pointed to in the way that Marie was harmed by SIM fraud, nor is there a discrete corporate asset that has been stolen, as in a website cloning case. Instead, the harm is diffuse and collective: it is the *legitimacy of the election result itself*, *public confidence in the democratic process*, and *social order and peace* that are damaged. Every citizen who voted, every candidate who stood, and the nation's broader political stability are all affected simultaneously, even though no individual has had money stolen or a personal account compromised.

This distinction matters practically as well as conceptually, because it changes what investigators must look for and how the harm is measured. In a crime against persons, investigators trace evidence back to how a specific individual's data, device, or account was compromised. In a crime against property, investigators trace ownership, authorisation, and asset value — who owns this code, this website, this data, and was it taken without permission for profit or harm. In a crime against society, by contrast, investigators must additionally grapple with questions of scale, public impact, and systemic integrity: how many polling stations show

anomalies, how widely did the disinformation spread, and what was its demonstrable effect on public behaviour and trust. The evidence base reflects this difference — rather than a single device or a single company's server logs, investigators draw on statistical analysis of election-wide datasets, social media network analysis to trace the spread of disinformation across thousands of accounts, and platform-level cooperation to understand coordinated inauthentic behaviour.

It is also worth noting that crimes against society often *intersect* with the other categories rather than existing in total isolation from them. The Ponzi scheme example, discussed further in Essay 2 below, illustrates this overlap clearly: a large-scale financial fraud scheme that defrauds thousands of members of the public simultaneously, damages public trust in financial systems and online platforms more broadly, and constitutes a property crime (theft of individual victims' money) at the same time as it constitutes a crime against society (erosion of trust in the digital economy as a whole, and the social harm caused by mass financial ruin within a community). Classification, therefore, is not always a matter of identifying a single "correct" category but of recognising which dimension of harm — individual, organisational/financial, or collective/societal — is *primary* in a given case, and election interference stands as society's purest example: an attack whose entire purpose and effect is to corrode the shared democratic fabric that no single victim could individually represent.

Essay 2 (17 marks)

Using the SUNRUN Ponzi scheme as a case study, explain how social media and mobile money platforms can be exploited to facilitate large-scale financial fraud against the public, and discuss the digital forensic methodology that would be applied to investigate such a case after the operators disappear.

The SUNRUN Ponzi scheme illustrates how the combination of social media's viral reach and mobile money's frictionless transaction infrastructure can be exploited to defraud large numbers of ordinary citizens at a scale and speed that would have been far harder to achieve through traditional, in-person fraud schemes. Understanding both the mechanics of exploitation and the forensic methodology needed to investigate such schemes after their operators vanish is essential to appreciating why these cases represent a distinctive and growing category of cybercrime against society and property simultaneously.

A scheme of this type typically begins with the operators establishing a seemingly legitimate investment opportunity — promising returns far above what any genuine investment vehicle could realistically sustain — and using social media platforms (Facebook, WhatsApp groups, Telegram channels) to advertise these returns aggressively. Social media is exploited in several specific ways: it allows the scheme's promoters to reach a very large audience cheaply and quickly, often using paid advertisements, influencer endorsements, or coordinated posting from multiple accounts to create an impression of widespread legitimacy and social proof; it allows for rapid, low-cost dissemination of testimonials (genuine or fabricated) from early "successful" investors, which is one of the most psychologically persuasive elements of any Ponzi scheme since it leverages trusted peer networks rather than impersonal advertising; and it allows the scheme to operate with minimal physical infrastructure or visible institutional presence, making it harder for sceptical observers or regulators to immediately identify red flags that might be obvious with a registered, storefront financial institution.

Mobile money platforms compound this vulnerability by providing the transactional backbone that makes the scheme operationally viable. Victims can deposit funds instantly via mobile money transfer without needing a bank account, in-person verification, or any of the friction that traditionally accompanied investment transactions; this lowers the barrier to participation dramatically, drawing in financially unsophisticated victims

who might never have engaged with a traditional bank-mediated investment scheme. Mobile money also enables the scheme operators to quickly disburse "returns" to early investors — which, critically, are not genuine investment profits but simply funds taken from newer investors — reinforcing the appearance of legitimacy and encouraging victims to recruit friends and family, further accelerating the scheme's viral growth through their own social networks. When the scheme inevitably becomes unsustainable (as all Ponzi structures eventually must, since they depend on an ever-growing pool of new contributors to pay earlier ones), mobile money's relative anonymity and the ease of withdrawing and disappearing with funds allows operators to vanish quickly, often before victims realise the scheme has collapsed.

Investigating such a case after the operators have disappeared presents distinctive forensic challenges, but a structured methodology can still yield substantial results. **Identification** begins once a critical mass of victim complaints establishes that a fraudulent scheme, rather than a failed but genuine investment, has occurred — distinguishing the two requires examining whether the promised returns were structurally impossible (a strong indicator of a Ponzi structure) and whether the operators' marketing claims were knowingly false. **Collection** focuses on gathering the digital trail left behind: mobile money transaction records showing the flow of funds from victims to the scheme's accounts and onward to operator-controlled accounts; social media account data, including posting history, advertisement records, and the network of accounts used to promote the scheme; and any website or app infrastructure used to track victim "investments," including domain registration records and hosting details. **Preservation** of this evidence is particularly urgent in such cases because social media accounts can be deleted by the operators or suspended by the platform itself, and mobile money records may be subject to retention policies that eventually purge older data — meaning investigators must move quickly to formally request preservation orders from telecom operators and social media platforms. **Examination and analysis** then involve using link-analysis tools such as Maltego to map the network of accounts, phone numbers, and mobile money wallets involved, tracing fund flows to identify the ultimate beneficiary accounts (which, even if the operators have fled, often cannot be fully concealed without leaving some traceable pattern, especially where funds were converted to cash through identifiable mobile money agents) and correlating the timing of account creation, advertisement posting, and fund collection to build a timeline establishing fraudulent intent from the outset. **Presentation** of findings must carefully document the total financial harm, the number of victims affected (supporting the "crime against society" framing through its sheer collective scale), and a clear evidentiary chain connecting the operators to the scheme's infrastructure and proceeds, even in their physical absence — since prosecution in absentia, asset recovery, or future apprehension may all depend on the strength of this documented trail.

Ultimately, the SUNRUN case demonstrates that the same features which make social media and mobile money valuable for legitimate financial inclusion and communication — low barriers to entry, rapid reach, minimal verification — are precisely what fraudsters exploit to defraud the public at scale, and that effective investigation after such schemes collapse depends on the rapid preservation of inherently perishable digital evidence before it disappears along with the perpetrators.

Essay 3 (16 marks)

"Disinformation and AI-generated deepfakes pose a greater threat to democratic societies than traditional hacking of electoral systems." Critically evaluate this statement, supporting your argument with reference to digital forensic evidence categories and investigation challenges.

This statement presents a genuinely contestable claim, and a critical evaluation requires weighing the distinctive harms, detectability, and investigability of each threat rather than assuming one is simply more "dangerous" in

the abstract. There are compelling arguments on both sides, and the strongest position is a qualified agreement: disinformation and deepfakes likely pose a *more diffuse and harder-to-remedy* threat to democratic societies, even though traditional hacking can cause more acute, immediately verifiable damage.

In favour of the statement, consider first the matter of **scale and virality**. A direct hack of an electoral database — for instance, the unauthorized alteration of vote tallies — is, once detected, a discrete and bounded problem: forensic investigators can examine server logs, identify unauthorized access timestamps, compare current data against backups or paper trail records, and in principle restore the correct figures with reasonable confidence once the intrusion is understood. By contrast, a viral deepfake video falsely depicting a candidate making inflammatory remarks can spread to hundreds of thousands of viewers within hours, across multiple platforms and via private messaging apps that are difficult to monitor, and even after the video is debunked or removed, the false impression it created persists in public consciousness — a phenomenon sometimes called the "continued influence effect," where corrections rarely fully undo the damage of an initial false belief. This asymmetry between the speed of harmful spread and the much slower speed of correction or evidentiary unwinding is a powerful argument for the greater societal danger of disinformation.

Second, consider **evidentiary and investigative difficulty**. Direct hacking of electoral systems typically leaves a well-understood category of digital forensic evidence: server access logs, intrusion detection alerts, database modification timestamps, and network traffic indicating unauthorized access — all categories that forensic investigators are well-trained and well-tooled to analyse using established methods (Splunk-based log analysis, network forensics with Wireshark, and so on). Disinformation and deepfakes, by contrast, present comparatively novel and difficult evidentiary challenges: establishing that a video is an AI-generated fabrication may require specialised forensic techniques to detect subtle artefacts in pixel patterns, audio waveforms, or metadata that AI generation tools leave behind, and these detection techniques continually lag behind rapidly improving generative AI capabilities. Furthermore, attributing responsibility for a deepfake's *creation and deliberate spread* is significantly harder than attributing a server intrusion, because the content may be reshared by thousands of seemingly independent accounts (some genuine, some coordinated bot networks), making it extremely difficult to trace back to an original malicious actor with the same confidence that an IP address or login credential trail might offer in a hacking case.

On the other hand, a case can be made that traditional hacking of electoral systems remains the more dangerous threat in certain respects, since a successful direct manipulation of vote counts strikes at the literal mechanical integrity of the democratic process — if undetected, it could directly and silently determine an election's outcome, which is arguably a more severe and irreversible harm than disinformation, which at least operates in the open and remains, in principle, contestable by counter-speech, fact-checking, and public scepticism. Hacking, in this view, is more dangerous specifically *because* it can be invisible and conclusive, whereas disinformation's visibility — however damaging — leaves room for societal correction mechanisms to operate, however imperfectly.

On balance, the more defensible critical position is that disinformation and deepfakes represent the *greater systemic and long-term threat to democratic societies*, not because any single incident is necessarily more damaging than a successful direct hack, but because the investigative and evidentiary tools available to detect, attribute, and correct disinformation lag significantly behind those available for detecting and forensically reconstructing traditional system intrusions, and because the social harm of disinformation — eroded trust, polarisation, and the difficulty of ever fully "undoing" a false belief once widely held — tends to outlast the technical remediation of a hacking incident. This suggests that forensic and regulatory capacity-building should increasingly prioritise developing robust deepfake detection and disinformation-network attribution methodologies alongside the more mature toolset already available for traditional electoral system hacking.

Essay 4 (17 marks)

Describe the five phases of digital forensic investigation methodology (identification, collection, preservation, analysis, presentation) and illustrate each phase using an election-related cybercrime scenario of your choice.

The digital forensic investigation methodology provides a structured framework that ensures evidence is gathered, processed, and presented in a way that is both technically sound and legally defensible. To illustrate each phase concretely, consider the scenario introduced earlier: following a Cameroonian presidential election, several polling stations report turnout exceeding registered voter numbers, while a coordinated wave of AI-generated deepfake content simultaneously spreads across social media, prompting the electoral commission to request a forensic investigation into suspected coordinated election interference.

Identification is the foundational phase in which investigators determine that an incident requiring investigation has actually occurred, and define its scope. In this scenario, identification begins when statisticians within the electoral commission notice the turnout anomalies during routine results verification, and simultaneously, the commission's communications team flags the viral deepfake content as a possible disinformation campaign. At this stage, investigators must determine: which polling stations show anomalous figures, what systems generated and transmitted those figures, which social media platforms and accounts are involved in spreading the deepfake content, and whether there is reasonable basis to suspect a coordinated rather than incidental cause. This phase essentially answers the question "what exactly are we investigating, and across which systems?"

Preservation follows immediately and addresses the urgent risk that critical evidence may be lost, overwritten, or deleted before it can be properly collected. In this scenario, preservation would involve formally requesting that the electoral commission's IT department freeze and back up the result-reporting database in its current state (preventing any further automated processing that might overwrite the anomalous records), and submitting urgent preservation requests to the relevant social media platforms to retain account data, posting histories, and any available metadata related to the deepfake video's upload and distribution before accounts can be deleted or content removed by the platforms themselves (which, ironically, might otherwise inadvertently destroy evidence in the course of routine content moderation).

Collection is the active gathering of the preserved evidence using forensically sound methods that do not compromise its integrity. For the turnout anomaly, this would involve obtaining certified exports of the full polling station results dataset, including timestamps of data entry and any subsequent modifications. For the deepfake content, collection would involve downloading the original video file together with its full metadata, capturing screenshots and timestamps of its spread across multiple platforms, and, where legally possible, obtaining backend account data (creation dates, IP login history, associated devices) from the social media platforms for the accounts most responsible for initial and viral distribution.

Analysis is the interpretive phase in which the investigator draws substantive conclusions from the collected evidence. For the turnout data, this would involve running statistical comparisons of each flagged polling station's reported turnout against its registered voter population and historical turnout patterns, identifying which anomalies are statistically extreme enough to indicate deliberate manipulation rather than simple clerical error. For the deepfake video, analysis would involve forensic examination of the file for AI-generation artefacts (inconsistent lighting, audio-visual lip-sync mismatches, telltale compression patterns associated with known generation tools) and a network analysis of the accounts that shared it, looking for signs of coordinated, non-

organic amplification such as near-simultaneous posting times, newly created accounts with no prior activity, or repeated use of identical caption text — all of which would support an inference of deliberate, organised interference rather than spontaneous public sharing.

Presentation is the culminating phase in which the investigator's findings are synthesised into a clear, structured report for the electoral commission, law enforcement, and potentially the courts. This report must lay out, in a logically organised and evidence-supported manner, which specific polling stations showed statistically anomalous results and the degree of confidence that this reflects manipulation rather than error; the technical basis for concluding the video was an AI-generated fabrication rather than authentic footage; and the evidentiary trail connecting the video's spread to a pattern of coordinated, rather than organic, social media activity. The report's findings would then inform whatever combination of legal, electoral, and public-communication responses the situation warrants — potentially including the annulment of results from specifically compromised polling stations, public debunking of the deepfake content with the forensic evidence as support, and referral of identified accounts for further investigation and possible prosecution.

This walkthrough demonstrates that even in a complex, multi-faceted incident combining statistical data manipulation and AI-generated disinformation, the same five-phase structure provides a coherent and adaptable framework, with each phase's specific activities shaped by the particular technical nature of the evidence involved while the underlying logic — define the scope, protect the evidence, gather it properly, interpret it rigorously, and communicate it clearly — remains constant.

Essay 5 (16 marks)

Discuss the challenges associated with digital evidence in crimes against society, with particular attention to jurisdiction, attribution, and the difficulty of tracing anonymous social media accounts.

Crimes against society, by their very nature, tend to present some of the most difficult evidentiary challenges in digital forensics, precisely because the harm they cause is diffuse, the actors involved are often numerous and geographically dispersed, and the platforms through which such crimes are committed are frequently designed — whether intentionally or not — to preserve user anonymity. Three interrelated challenges deserve particular attention: jurisdiction, attribution, and the specific difficulty of tracing anonymous social media accounts.

Jurisdiction presents a foundational obstacle because cybercrimes against society, such as election interference through disinformation, very rarely respect national borders. The servers hosting a deepfake video, the social media platform's corporate headquarters, the accounts responsible for initial upload, and the ultimate victim population (Cameroonian voters, in the election interference example) may all be located in entirely different countries, each governed by different legal systems with different definitions of what constitutes illegal disinformation, different evidentiary standards, and different levels of willingness to cooperate with a Cameroonian investigation. A request for data from a major social media platform headquartered in the United States, for instance, may need to satisfy that country's legal standards for data disclosure, which can be slow, may require formal mutual legal assistance treaty (MLAT) processes, and may simply be denied if the platform judges the request insufficiently specific or the underlying conduct not clearly illegal under the laws of the country where the platform is based. This jurisdictional fragmentation means that even when investigators have identified precisely what evidence they need, actually obtaining it can take months, by which point the political moment the disinformation was designed to influence — an election, for instance — has already passed, rendering the eventual evidence more useful for accountability than for timely intervention.

Attribution is a closely related but distinct challenge: even once relevant data is obtained, establishing *who* is genuinely responsible for orchestrating a coordinated disinformation campaign or interference effort is technically difficult. Sophisticated actors typically route their activity through virtual private networks (VPNs), proxy servers, or compromised third-party accounts specifically to obscure their true location and identity. A deepfake video might be uploaded from an account that itself shows no clear link to any real-world identity, hosted on infrastructure rented anonymously using cryptocurrency, and amplified by a network of accounts that, on the surface, appear to be ordinary citizens engaging in normal political discussion but are in fact coordinated bot or troll accounts. Distinguishing genuine grassroots public reaction from deliberately orchestrated amplification requires sophisticated network analysis — examining account creation patterns, posting time correlations, linguistic similarities across supposedly unrelated accounts, and other behavioural fingerprints — and even with such analysis, definitively proving a *specific* responsible individual or organisation (as opposed to merely demonstrating that *some* coordinated effort occurred) often remains elusive, particularly when the operation has been deliberately designed with attribution-resistance in mind.

Tracing anonymous social media accounts compounds both of the above challenges at a more granular, operational level. Many platforms allow account creation with minimal identity verification, meaning the accounts most actively involved in spreading disinformation may have no verifiable connection to a real person at the platform level at all. Even where a platform retains backend data — IP addresses used at signup or login, device fingerprints, payment information for any associated advertising — this data is typically only accessible to investigators through the platform's voluntary cooperation or formal legal compulsion, both of which loop back into the jurisdictional challenges discussed above. Furthermore, sophisticated actors may use IP-masking technology specifically to defeat this kind of backend tracing, meaning that even successful legal access to platform data may yield IP addresses that themselves lead only to VPN exit nodes rather than any real, traceable location.

Taken together, these three challenges mean that investigations into crimes against society frequently produce strong evidence that *something* improper occurred — clear statistical anomalies, demonstrably fabricated content, identifiable patterns of coordinated amplification — without necessarily producing the kind of individually attributable, jurisdictionally actionable evidence needed to bring specific perpetrators to justice. This has led to growing emphasis, in international cybersecurity policy discussions, on strengthening cross-border cooperation frameworks, encouraging platform transparency commitments (such as mandatory disclosure of coordinated inauthentic behaviour), and developing shared international norms around state and non-state responsibility for cyber-enabled interference in democratic processes — recognising that purely domestic forensic capability, however skilled, cannot fully resolve challenges that are inherently transnational in nature.

SET 3 — Crimes Against Property

Essay 1 (17 marks)

Discuss the concept of cybercrime against property, with reference to website cloning as a specific example. Explain the various forms of digital property that can be infringed and the objectives attackers typically pursue.

Cybercrime against property encompasses digital offences in which the principal target is a tangible or intangible asset — financial resources, intellectual property, digital infrastructure, or organisational data — rather than an individual's personal identity or wellbeing (as in crimes against persons) or the collective

institutions of society (as in crimes against society). In the digital era, "property" has expanded well beyond physical objects to include source code, brand identity, domain names, proprietary content, and the trust relationship between an organisation and its customers — all of which can be stolen, copied, or exploited without ever touching a physical object.

Website cloning is among the clearest and most instructive examples of cybercrime against property, precisely because it demonstrates how multiple distinct forms of digital property can be infringed within a single criminal act. The case of the fraudulent duplication of Cameroon's official biometric passport pre-enrollment website illustrates this well: shortly after the legitimate "passcam.cm" portal launched, fake domains such as "passcam.atlassian.net" appeared, designed to closely mimic the original in appearance and function, deceiving citizens into paying enrollment fees to a fraudulent site that bore no genuine connection to the passport process. Several distinct categories of property were infringed in this single act. First, **intellectual property** was stolen in the form of the website's unique source code, design layout, graphics, and written content, all of which legally belong to the Cameroonian government and were reproduced without authorisation. Second, **brand identity and domain reputation** were exploited: the cloned site derived its entire deceptive power from borrowing the trust and recognition that citizens had already placed in the legitimate government passport system, effectively stealing the *reputational value* the real site had built. Third, **financial property belonging to individual victims** was targeted, since citizens who paid fees to the fraudulent site lost real money under false pretences — illustrating how property crimes against an organisation (the cloned site, infringing the government's digital assets) and against persons (the victims who paid fraudulent fees) frequently occur simultaneously within a single scheme. Fourth, in a broader sense, **the integrity and authority of the legitimate platform itself suffered damage** — described in the forensic literature as a form of cyber vandalism against the original site's standing, since the existence of a convincing clone undermines public confidence in the authenticity of official digital services going forward, even after the fraudulent site is taken down.

The objectives that attackers typically pursue through website cloning are similarly varied but tend to cluster around a few recurring goals. The most direct objective is **credential and financial information theft**: cloned sites frequently include fraudulent login pages or payment portals designed specifically to harvest usernames, passwords, or banking details from unsuspecting users who believe they are interacting with the genuine service. A second objective is **direct financial fraud through fee collection**, as in the passport case, where victims are simply persuaded to pay money for a service that the fraudulent site has no ability or intention to actually deliver. A third objective is the **distribution of malware**, where a cloned site may be used as a vector to infect visitors' devices, either through malicious downloads disguised as legitimate documents or through drive-by exploitation of browser vulnerabilities. A fourth and more subtle objective is **reputational exploitation** for purposes beyond immediate financial gain — for instance, a cloned site might be used to spread misinformation under the guise of an authoritative source, leveraging the original organisation's credibility to lend false legitimacy to fraudulent claims.

What makes website cloning a particularly clear illustration of cybercrime against property is that it requires no breach of any system's internal security at all — the attacker need not hack the original website's servers, since cloning relies entirely on *external, public-facing replication* of what any visitor can already see and copy. This distinguishes it sharply from other property crimes such as database or backend theft, which do require unauthorized system access. It is precisely this characteristic — that valuable property can be stolen simply by copying what is visibly presented to the public, without ever defeating any technical security control — that makes website cloning such a deceptively low-barrier yet highly damaging form of cybercrime against property,

and which underscores why legal protections (under both cybercrime law and intellectual property law) and rapid detection capabilities are essential complements to traditional technical security measures.

Essay 2 (17 marks)

Using the Cameroon passport pre-enrollment website cloning incident as a case study, describe the full digital forensic investigation procedure — from identification of the cloned website through to attribution and evidence preservation — explaining the tools used at each stage.

The investigation of the Cameroon passport pre-enrollment website cloning incident illustrates a complete digital forensic procedure tailored specifically to property crimes involving website duplication, moving systematically from initial discovery through to evidence preservation suitable for prosecution.

The investigation begins with **identification of the cloned website**, the activity of confirming that a suspected fraudulent site is indeed an unauthorized duplicate of the legitimate "passcam.cm" portal rather than, for instance, an unrelated coincidentally similar site. Investigators would systematically compare the original and suspected cloned website across several dimensions: overall layout and visual design, embedded images and logos, source code structure, and the domain name itself. At this stage, **Httrack** would be used to download a complete local copy of both the legitimate and suspected fraudulent websites, enabling side-by-side offline comparison without repeatedly querying the live (and potentially still-active, harmful) fraudulent site, while **browser developer tools** would be used to inspect the live HTML, CSS, and JavaScript of both sites directly in the browser, allowing investigators to quickly spot identical code structures, comments, or embedded tracking scripts that suggest direct copying rather than independent development. The evidence collected at this stage — screenshots, full HTML/CSS/JavaScript files, and the respective URLs — establishes the foundational basis for the rest of the investigation.

The investigation then proceeds to **domain and DNS investigation**, aimed at determining who registered the fraudulent domain ("passcam.atlassian.net") and where it is hosted. **ICANN Lookup** would be the primary tool here, used to perform a WHOIS lookup that retrieves the domain's registration details, including the registrant's name (if not privacy-shielded), the registration date, and the associated registrar. Investigators would specifically examine whether the registration date closely preceded the launch of the genuine "passcam.cm" portal (suggesting the clone was prepared in anticipation of the official launch) and would cross-reference DNS records to identify the hosting provider and IP address serving the fraudulent domain — information critical both for understanding the scale of the operation and for potentially requesting takedown action from the hosting provider.

Source code and content analysis follows, in which the original site's code is systematically compared against the suspected clone's code. **WinMerge** would be the principal tool at this stage, performing detailed line-by-line comparison of the HTML, CSS, and JavaScript files collected earlier, automatically highlighting identical or near-identical code blocks, structural patterns, and even retained developer comments that may have been inadvertently copied along with functional code — all of which constitute strong, objectively verifiable evidence of direct duplication rather than coincidental similarity.

The investigation then turns to **malware and phishing analysis**, examining whether the cloned site contains malicious functionality beyond simple visual deception — particularly relevant given that victims were directed to pay fees through the fraudulent portal. **VirusTotal** would be used to scan the fraudulent site's URL and any downloadable content it offers, checking whether security vendors and threat intelligence databases have already flagged the domain or associated files as malicious, while manual inspection of the site's payment form

handlers would reveal whether submitted financial details were being transmitted to attacker-controlled infrastructure rather than any legitimate payment processor.

Log analysis and digital forensics would then be applied to any server-side evidence available — for instance, if the hosting provider cooperates with the investigation or if a forensic copy of server logs can be obtained through legal process. **Splunk** would be used to analyse these logs, searching for patterns such as the volume and timing of visits to the fraudulent site, the geographic distribution of victims (helpful in understanding the scale of public harm), and any technical fingerprints (server software versions, configuration patterns) that might later help link this incident to other cloning operations by the same actor.

Attribution and tracing the attacker represents the most challenging phase, attempting to move from technical evidence toward identifying the actual person or group responsible. **Maltego** would be employed here to perform open-source intelligence (OSINT) and link analysis, mapping relationships between the fraudulent domain's registration details, any associated email addresses, hosting account information, and potentially connected social media or financial accounts used to receive victim payments — building a network graph that may reveal connections obscured by any single piece of evidence in isolation.

Finally, **evidence preservation** ensures that everything gathered throughout the investigation remains legally usable. **Autopsy** would be used to create and analyse forensic images of any seized devices or server data obtained during the investigation, while every piece of digital evidence collected — website code, WHOIS records, log files, OSINT findings — would be hashed (using algorithms such as MD5 or SHA-1) to mathematically prove its integrity, with a meticulously documented chain of custody recording every individual who accessed or analysed the evidence at every stage, ensuring the entire investigative trail remains defensible if the case proceeds to prosecution.

This case demonstrates how a property crime as conceptually simple as "copying a website" in fact requires a sophisticated, multi-tool, multi-stage forensic methodology to properly identify, analyse, attribute, and legally substantiate — reflecting the genuine technical and investigative complexity that underlies even seemingly straightforward cybercrimes against property.

Essay 3 (16 marks)

Compare and contrast the evidentiary challenges involved in investigating website cloning versus investigating database or backend theft. In your answer, discuss the specific tools and evidence sources relevant to each.

Website cloning and database or backend theft are both forms of cybercrime against property, yet they present meaningfully different evidentiary profiles, investigative entry points, and challenges, which is instructive to compare directly.

Website cloning, as exemplified by the Cameroon passport pre-enrollment case, is distinguished by the fact that the core evidence is largely **external and publicly visible**. Because a cloned website must, by definition, present itself to the public in order to deceive visitors, investigators can begin gathering substantial evidence — the cloned site's design, source code, domain registration, and even any malicious functionality — without needing privileged access to any victim organisation's internal systems at all. Tools such as **Htttrack** (for capturing complete copies of both sites for comparison), **WinMerge** (for source code comparison), and **ICANN Lookup** (for domain registration details) can all be applied directly to publicly accessible resources. This gives website cloning investigations a relatively *fast and low-barrier initial evidentiary phase*: an investigator can

confirm, often within hours, that cloning has occurred and gather strong supporting evidence simply by examining what is visible to any internet user. The principal evidentiary challenge in cloning cases instead arises *later*, at the attribution stage — once the visible evidence of cloning is well-established, identifying the actual person or organisation behind the fraudulent domain requires deeper OSINT work (using tools like Maltego), and this stage is often complicated by anonymous domain registration services, privacy-shielded WHOIS records, and the use of offshore or loosely regulated hosting providers that may not cooperate readily with investigation requests.

Database or backend theft, by contrast, presents an almost inverse evidentiary profile. Because backend systems — the actual servers, databases, and internal infrastructure of an organisation — are not, by design, visible or accessible to the public, the **initial evidentiary phase is typically the harder and slower stage**: investigators cannot simply browse and compare what is publicly visible, but must instead rely on the victim organisation's cooperation to access server logs (analysed via tools such as **Splunk**), database backup files, SQL dumps, and any intrusion detection alerts that may have flagged unauthorized access. This dependency on the victim organisation's own internal logging and monitoring infrastructure introduces a significant evidentiary risk: if the organisation's logging practices were inadequate, or if logs were not retained for a sufficient period before the breach was discovered, crucial evidence may simply not exist by the time investigators arrive — a problem that is far less likely in cloning cases, where the cloned site itself typically remains observable for some time, and where domain registration records are maintained by third-party registrars largely independent of the victim's own diligence. Once internal log and database evidence is obtained, however, the *forensic interpretation* of backend theft can in some respects be more directly conclusive than cloning evidence, since matching database schemas, identical customer data structures, or duplicate product inventories provide very strong, almost mathematically precise evidence of direct copying — comparable in evidentiary strength to source code comparison in cloning cases, but applied to structured data rather than visual/code presentation.

A further point of contrast lies in the **scale and nature of harm typically associated with each**. Website cloning, as in the passport case, often causes harm primarily through *deception of end users* who interact with the fraudulent site directly — meaning the population of affected victims may be large but each individual instance of harm (a fee paid, credentials entered) is relatively discrete and traceable to a specific interaction. Backend theft, by contrast, often results in the wholesale compromise of an organisation's entire customer database in a single event, meaning the evidentiary task shifts from "did copying occur" (frequently obvious once backend access is confirmed) toward "what is the full scope of data exfiltrated and which specific individuals are affected" — a scoping challenge that can be extremely labour-intensive, particularly with large databases, and which has significant implications for an organisation's legal disclosure obligations to affected customers.

In summary, website cloning investigations tend to move quickly through an accessible, public-facing evidentiary phase but face their greatest challenge at attribution, while database/backend theft investigations face their greatest challenge earlier, in simply obtaining sufficient internal evidence from a cooperating (and adequately logging) victim organisation, but once obtained, that evidence often more directly and conclusively establishes the fact and scope of the theft. Effective investigators must therefore be prepared to apply substantially different initial strategies — public OSINT-driven analysis for cloning, and victim-cooperation-dependent internal forensics for backend theft — despite both ultimately falling under the same broad category of cybercrime against property.

Essay 4 (17 marks)

"Domain name similarity and typo-squatting are the easiest indicators to detect, yet the hardest to prosecute." Critically discuss this statement with reference to jurisdiction, attribution, and the legal requirements for proving intent to deceive.

This statement captures a genuine and important tension in the investigation and prosecution of website cloning and related domain-based cybercrimes against property, and a critical discussion reveals why technical detectability and legal prosecutability can diverge so sharply in this particular area.

The claim that typo-squatting and domain similarity are **easy to detect** is well-supported. Identifying that "passcam.atlassian.net" is suspiciously similar to "passcam.cm," or that "afmart-cm.shop" closely mimics "afmart.cm," requires no sophisticated technical capability — a simple visual comparison, or at most a basic WHOIS lookup using a tool like **ICANN Lookup**, can confirm the existence of a deceptively similar domain within minutes. This stands in marked contrast to many other forms of cybercrime, where even detecting that an offence has occurred at all (let alone identifying the specific technical mechanism) can require extensive forensic analysis. The visual and structural nature of domain similarity makes it almost uniquely "self-evident" as a category of suspicious activity, which is precisely why typo-squatting detection can often be partially automated, with organisations proactively monitoring for newly registered domains that closely resemble their own as a routine brand-protection measure.

However, the statement's second claim — that such cases are simultaneously **among the hardest to prosecute** — is equally well-supported, for several converging reasons. First, **jurisdiction** poses a major obstacle: domain registrars and hosting providers operate globally, and a fraudulent domain targeting Cameroonian citizens, as in the passport cloning case, may be registered through a registrar based in an entirely different country, with servers hosted in yet another jurisdiction. Compelling such registrars or hosting providers to disclose registrant information, take down the fraudulent domain, or preserve evidence for a Cameroonian investigation often requires navigating that country's own legal processes, which may be slow, may demand a level of formal evidentiary specificity that is difficult to provide at an early investigative stage, or may simply not prioritise a request from a foreign jurisdiction highly.

Second, **attribution** remains genuinely difficult even after jurisdiction is successfully navigated, because domain registration can frequently be conducted through privacy-shielding services that conceal the true registrant's identity behind a proxy, and payment for domain registration and hosting can be made through cryptocurrency or prepaid services that are themselves difficult to trace to a real individual. Even where some identifying information is obtained, sophisticated actors may have used entirely fabricated or stolen identity details during registration, meaning that what appears to be attribution evidence may in fact lead investigators to an innocent third party whose identity was itself misappropriated — adding a further layer of complexity reminiscent of the identity-theft dynamics seen in crimes against persons.

Third, and perhaps most legally significant, is the requirement to **prove intent to deceive** — a requirement central to Cameroon's framework for distinguishing criminal cloning from legitimate activity, since not all website duplication is illegal (a developer testing their own clone for legitimate purposes commits no offence). Proving intent typically requires demonstrating that the domain was registered specifically to mimic the original with the deliberate purpose of misleading visitors, rather than for some innocent or coincidental reason. While in egregious cases like the passport portal cloning this intent may seem self-evident given the near-perfect mimicry and the collection of fraudulent fees, defence arguments can still contest specific elements — for instance, arguing that visual similarity alone does not prove the *registrant* (if a different individual is found to be merely the hosting reseller or an unwitting intermediary) had the requisite criminal intent, particularly if attribution evidence is itself circumstantial rather than direct. Courts may require a higher evidentiary threshold

connecting a specific identified individual not just to the domain's existence, but to its deliberate creation and operation with deceptive purpose — a standard that is considerably harder to satisfy than the comparatively simple technical observation that two domains look alike.

The cumulative effect of these three factors — slow and uncertain cross-border legal cooperation, genuine difficulty in identifying a real, provable human actor behind deliberately obscured registration details, and the legal demand for proof of subjective criminal intent rather than mere technical similarity — explains why typo-squatting cases so often stall between an early, confident technical identification and a much less certain, frequently abandoned or indefinitely delayed prosecution. This gap has practical implications: it suggests that domestic remedies such as rapid administrative domain takedown procedures (which do not require the full evidentiary burden of criminal prosecution) and stronger international registrar cooperation agreements may, in practice, offer more realistic protection for victims than pursuing full criminal prosecution to conclusion in every case, even though prosecution remains an important deterrent and accountability mechanism where it can be successfully pursued.

Essay 5 (16 marks)

Discuss the role of evidence preservation (hashing, forensic imaging, chain of custody) in cybercrime-against-property investigations, and discuss the consequences of failing to properly preserve such evidence.

Evidence preservation forms the structural backbone of any successful cybercrime-against-property investigation, since even the most technically sophisticated and conclusive analysis is rendered worthless if the underlying evidence cannot be shown, beyond reasonable challenge, to be authentic and unaltered. Three specific preservation mechanisms — hashing, forensic imaging, and chain of custody — work together to provide this assurance, and understanding both their function and the consequences of their absence is essential to appreciating why preservation receives such emphasis throughout digital forensic methodology.

Hashing refers to the process of running a piece of digital evidence (a file, a complete disk image, a captured webpage) through a cryptographic hash algorithm, such as MD5 or SHA-1, which produces a unique, fixed-length string of characters mathematically derived from the evidence's exact content. Any alteration to the underlying evidence — even a single changed character or bit — will produce a completely different hash value, making hashing an extremely reliable mechanism for proving, at any later point, that a piece of evidence remains identical to its state at the moment of collection. In a website cloning investigation, for instance, investigators would compute hash values for the downloaded copies of both the original and suspected cloned websites immediately upon collection (using Httrack), and again at the point the evidence is presented in court, allowing them to demonstrably prove that no tampering — whether accidental or deliberate — occurred in the intervening period.

Forensic imaging refers to the creation of an exact, complete, bit-for-bit copy of a storage medium or digital resource, performed in a manner (typically using write-blocking hardware or software) that guarantees the original source is never altered during the copying process itself. This is critical because the very act of accessing or analysing digital evidence can, if done carelessly, alter timestamps, modify file metadata, or otherwise change the evidence in ways that compromise its evidentiary value. By working exclusively from a verified forensic image — using tools such as **FTK Imager** for general digital evidence or **Autopsy** for subsequent file system analysis — investigators ensure that the original evidence (a seized server's hard drive, for instance, in a

backend database theft case) remains untouched and available as an unimpeachable reference point, while all analytical work proceeds on the verified copy.

Chain of custody is the continuous, meticulously documented record tracking every individual who has had access to a piece of evidence from the moment of its collection through to its presentation in court, including precisely when, how, and why each transfer or access event occurred. In a property crime investigation involving, for example, a seized server implicated in hosting a cloned website, chain of custody documentation would record exactly which officer seized the device, the precise time and location of seizure, every subsequent instance in which the device or its forensic image was accessed by an analyst, and how the device was stored (in what secured location, accessible to whom) at every point in between.

The consequences of failing to properly preserve evidence through these mechanisms can be severe and, in many cases, fatal to an otherwise strong case. If hashing is not performed, or if a hash mismatch is discovered between the original collection point and a later presentation in court, defence counsel can credibly argue that the evidence may have been altered, tampered with, or fabricated at some point — and courts, quite reasonably, apply a precautionary principle that favours excluding evidence whose integrity cannot be verified, regardless of how compelling its content might otherwise appear. If forensic imaging is skipped or performed improperly — for instance, if an investigator directly browses files on an original seized device rather than working from a verified image — the investigator risks inadvertently altering file access timestamps or other metadata that might themselves have been evidentially significant (for example, in establishing exactly when a cloned website's files were created relative to the original), potentially destroying the very evidence the investigation sought to establish. If chain of custody documentation is incomplete or contains unexplained gaps — a period, for instance, where a seized device was accessible to personnel not properly authorised or logged — defence counsel can argue that the evidence may have been tampered with during that unaccounted-for period, and courts may exclude the evidence on this basis alone, entirely independent of whether tampering can be affirmatively proven to have actually occurred; the mere *possibility*, created by inadequate documentation, is often sufficient grounds for exclusion under common evidentiary standards.

The cumulative lesson is that technical investigative skill and legal evidentiary survivability are not the same thing, and that a property crime investigation which is technically excellent but procedurally careless in its preservation practices can still collapse entirely upon reaching court. This is why digital forensic training places preservation procedures on equal footing with analytical technique, and why experienced investigators treat hashing, imaging, and custody documentation not as bureaucratic afterthoughts but as integral, non-negotiable components of the investigative process from its very first moment.

SET 4 — Crimes Against Government & General Forensic Principles

Essay 1 (17 marks)

Describe the DFRWS investigative paradigm (identification, preservation, collection, examination, analysis, presentation) in detail, and explain why each phase is essential to ensuring evidence is admissible in a court of law.

The Digital Forensics Research Workshop (DFRWS) investigative paradigm provides the foundational structure underlying virtually all professional digital forensic practice, organising the investigative process into six distinct phases — identification, preservation, collection, examination, analysis, and presentation — each of which performs a specific function that is essential, individually and cumulatively, to ensuring that the resulting evidence withstands legal scrutiny.

Identification is the initial phase in which investigators recognise that a security incident or crime has occurred and determine its scope: which systems, devices, accounts, or data sources are potentially relevant. This phase is essential because an investigation that begins without a clearly defined scope risks two opposite failures — either overlooking critical evidence sources that fall outside an overly narrow initial assessment, or wasting limited investigative resources and time examining systems that have no genuine relevance to the incident, potentially delaying the investigation past the point where time-sensitive evidence (such as volatile system memory or rolling log files) remains available. Legally, a clearly documented identification phase also helps establish that the subsequent investigation was conducted with a defensible, principled basis rather than an arbitrary or unjustified search, which can matter if the legality of the investigation's scope is later challenged.

Preservation follows immediately, focused on ensuring that identified evidence is not lost, altered, or destroyed before it can be properly collected — for instance, by isolating an affected system from the network to prevent further intrusion or remote tampering, placing a mobile device in airplane mode or a signal-blocking enclosure to prevent remote wiping, or issuing formal preservation requests to third parties (such as telecom operators or social media platforms) whose data retention policies might otherwise result in automatic deletion. This phase is essential to admissibility because digital evidence is inherently volatile and easily altered; if evidence is allowed to degrade, be overwritten, or be tampered with (even inadvertently) before formal collection, no subsequent analytical rigour can recover its original evidentiary value, and any findings based on already-compromised evidence become vulnerable to challenge on the basis that the evidence presented may not accurately reflect the original state of affairs.

Collection is the active, methodologically rigorous gathering of preserved evidence, typically performed using forensically sound techniques such as write-blocking hardware (to prevent any modification of original storage media during copying) and verified imaging tools. This phase is essential because the *method* by which evidence is collected directly determines whether it can later be shown to be an accurate, unaltered representation of the original source — courts generally require demonstration that proper forensic collection methods were used, since collection performed carelessly (for example, directly copying files using standard operating system commands rather than a verified forensic imaging tool) can alter metadata or timestamps in ways that undermine the evidence's reliability.

Examination involves the technical processing of collected evidence to surface potentially relevant material — for instance, running keyword searches across a disk image, recovering deleted files, or extracting specific categories of data (browser history, transaction records, communication logs) relevant to the investigation's scope. This phase is essential because raw collected evidence (an entire disk image, for example) is typically far too voluminous and unstructured to be directly useful; examination transforms this raw material into a more manageable and relevant subset, and the methods used must themselves be sound and properly documented, since courts may require investigators to explain and justify their examination methodology if challenged on whether relevant evidence might have been overlooked or whether the examination process itself introduced any distortion.

Analysis is the interpretive phase in which the investigator draws substantive conclusions from the examined evidence — correlating timestamps, identifying patterns, reconstructing sequences of events, and forming a

coherent narrative of what occurred and how. This phase is essential to admissibility because raw evidence alone, however well preserved and collected, does not constitute a legal case; it is the investigator's analytical interpretation, transparently reasoned and methodologically sound, that connects evidence to legal conclusions, and this analysis itself must be defensible under cross-examination, since opposing counsel will often seek to challenge not the existence of evidence but the investigator's interpretation of what that evidence demonstrates.

Presentation is the final phase, in which findings are compiled into a clear, well-organised report and, frequently, oral testimony suitable for a non-technical legal audience including judges, lawyers, and juries. This phase is essential because even flawless technical work has no legal value if it cannot be communicated in a manner that allows the court to understand and evaluate it; presentation must accurately reflect the underlying findings without overstatement or omission, since any perceived exaggeration or selective presentation can undermine the investigator's credibility and, by extension, the weight given to otherwise sound evidence.

Together, these six phases form an interlocking chain in which each phase's integrity depends on the proper execution of the phases before it: flawed identification undermines preservation by failing to protect the right evidence; flawed preservation undermines collection by allowing evidence to degrade before it can be gathered; and so on through to presentation. This is why digital forensic professionals treat the paradigm not as a flexible suggestion but as a rigorous discipline, recognising that admissibility in court is not achieved at the presentation stage alone but is, in fact, the cumulative product of careful, well-documented practice at every preceding phase.

Essay 2 (17 marks)

Discuss disk imaging as a forensic technique. Explain the tools used to create forensic images, the importance of hash verification, and the risks of failing to image a device correctly before analysis.

Disk imaging is one of the most fundamental techniques in digital forensics, referring to the process of creating a complete, exact, bit-for-bit copy of a storage medium — a hard drive, solid-state drive, USB device, or mobile device's storage — in a manner that preserves every element of the original data, including not only visible files but also deleted files, file system metadata, unallocated space, and other artefacts that standard file-copying methods would not capture. Understanding why this technique is so central to forensic practice requires examining both how it is properly performed and what is lost or compromised when it is not.

The core principle underlying disk imaging is that **the original evidence must never be directly analysed or modified**; instead, all forensic work is performed exclusively on a verified copy, leaving the original in its pristine, as-seized state, available as an unimpeachable reference point should any question later arise about the accuracy of the working copy. This is achieved through specialised tools designed specifically for forensically sound copying. **FTK Imager** is among the most widely used tools for this purpose, capable of creating a complete image of a storage device while computing cryptographic hash values of both the source and the resulting image simultaneously, allowing immediate verification that the two match exactly. The imaging process typically employs **write-blocking** hardware or software, which physically or logically prevents any write operation from reaching the original device during the imaging process — this is critical because even the act of simply connecting a storage device to a standard computer system, without write-blocking, can trigger automatic operating system processes (such as indexing or metadata updates) that inadvertently alter the original evidence, however minor such changes might seem.

The **importance of hash verification** in this process cannot be overstated. Once an image has been created, the investigator computes a cryptographic hash (commonly MD5 or SHA-1, sometimes both for added assurance) of both the original device and the newly created image. If these hash values match exactly, this provides strong mathematical proof that the image is a perfect, unaltered replica of the original — a claim that is vastly more persuasive in court than a simple assertion that "the copy was made carefully." Crucially, this hash verification is typically repeated at multiple points throughout the investigation: immediately after imaging, before any analysis begins, and again before the evidence is presented in court, creating a continuous, verifiable record that the evidence has remained unaltered throughout the entire investigative process, regardless of how much time has elapsed or how many different analysts may have worked with the image.

Once a verified image exists, further analysis can proceed using tools such as **Autopsy** (built on The Sleuth Kit), which can parse the image's file system, recover deleted files that remain physically present in unallocated space but have had their file system pointers removed, reconstruct timelines of file creation and modification, and search across the entire image for specific keywords or file types relevant to the investigation — all performed safely against the image rather than the original device.

The **risks of failing to properly image a device before analysis** are substantial and can be fatal to an investigation's legal viability. If an investigator instead works directly on the original device — for instance, simply browsing its file system using standard tools to "take a quick look" before deciding whether formal imaging is warranted — several serious problems can arise. First, the very act of browsing can alter file access timestamps, which might themselves have held evidentiary significance (for example, establishing precisely when a suspect last accessed a particular file). Second, if the device contains any automated processes (operating system updates, scheduled tasks, antivirus scans) that activate upon being powered on or connected, evidence could be inadvertently modified or even deleted before any forensic safeguard is in place. Third, and most seriously from a legal perspective, any subsequent attempt to demonstrate the integrity of evidence gathered from an improperly handled original device becomes essentially impossible — there is no "original, untouched" reference point against which to verify that nothing was altered, since the original itself was the device that was directly manipulated. This exposes the entire body of evidence to a defence challenge arguing that the evidence cannot be trusted to accurately represent its state at the time of seizure, a challenge that, once raised credibly, often results in evidence being excluded entirely regardless of how compelling its substantive content might be.

Disk imaging, hash verification, and the disciplined refusal to work directly on original evidence thus together constitute not merely good practice but the essential technical foundation upon which the legal credibility of digital forensic findings depends; skipping or shortcutting this foundational step, however time-pressured an investigation might be, risks undermining the entire evidentiary value of an otherwise well-conducted investigation.

Essay 3 (16 marks)

Compare the four broad categories of cybercrime — against persons, against society, against government, and against property — using one real or hypothetical Cameroonian example for each category. Discuss how the investigative approach differs across categories.

The four broad categories of cybercrime — against persons, against society, against government, and against property — provide a useful classificatory framework for understanding both the nature of harm involved in different cyber offences and the distinct investigative approaches each category typically demands. Examining

one illustrative Cameroonian example per category reveals both the conceptual distinctions and the practical differences in investigative methodology.

Crimes against persons are illustrated by the case of SIM card registration fraud, in which Marie's national identity document was misused to fraudulently register a SIM card subsequently used to drain her mobile money account. The harm here is intensely personal: financial loss, violation of identity, and psychological distress experienced by a single, clearly identifiable victim. The investigative approach for this category centres on **individual device and account forensics**: extracting data from the suspect's phone using tools like Cellebrite UFED, examining the specific registration and transaction records tied to the victim's identity, and tracing a relatively contained, person-to-person chain of events from the fraudulent registration through to the financial withdrawal.

Crimes against society are illustrated by election interference through coordinated disinformation and statistical manipulation of turnout figures, where the harm is diffuse and collective — affecting public trust in democratic institutions and national stability rather than any single victim's personal interests. The investigative approach here shifts dramatically toward **scale-oriented and network-based analysis**: rather than tracing a single chain of events affecting one person, investigators must apply statistical anomaly detection across entire election-wide datasets, conduct social network analysis to trace the spread of disinformation across potentially thousands of accounts, and grapple with platform-level cooperation and cross-border jurisdictional challenges that rarely arise to the same degree in person-focused cases.

Crimes against government are illustrated by a hypothetical spear phishing attack targeting the administrative email system of a state-owned institution, such as Camtel, aimed at gaining unauthorized access to government communications or infrastructure control systems. The harm here is institutional and potentially national-security-relevant, since compromised government systems can affect public service delivery, national infrastructure, and state secrets. The investigative approach for this category emphasises **infrastructure and network-level forensics combined with national security coordination**: investigators must analyse email authentication failures and network intrusion patterns (using tools like Wireshark and Splunk), assess the potential scope of compromised government systems, and coordinate closely with specialised national agencies (such as ANTIC) given the heightened sensitivity and potential national security implications of government system compromise, which typically involves a level of inter-agency coordination not required in purely individual or even society-wide civilian cases.

Crimes against property are illustrated by the cloning of the Cameroon passport pre-enrollment website, where the harm targets specific digital assets — the government's intellectual property in its website design and code, its brand reputation, and the financial property of individual citizens deceived into paying fraudulent fees. The investigative approach here centres on **comparative analysis and attribution-focused OSINT**: comparing the original and cloned website's source code and design (using tools like WinMerge and Httrack), investigating domain registration records (using ICANN Lookup), and conducting link analysis (using Maltego) to attempt to identify the individuals or groups responsible for the fraudulent operation.

Comparing across these four examples reveals a clear pattern: as the category of harm shifts from the individual (persons) toward the institutional and collective (government and society), the investigative approach correspondingly shifts from device-and-account-level forensics toward network-level, statistical, and inter-agency coordinated analysis; while property crimes occupy something of a middle position, often beginning with relatively accessible, externally observable evidence (as in website cloning) before requiring the same kind of attribution-focused OSINT work that characterises the more diffuse society and government categories. This suggests that effective forensic training cannot rely on a single uniform methodology but must equip

investigators with a flexible toolkit spanning individual device forensics, network and log analysis, statistical anomaly detection, and OSINT-based attribution techniques, deploying the appropriate combination based on which category of harm a given case most centrally represents.

Essay 4 (17 marks)

Discuss the skills, qualifications, and ethical standards required of a digital forensic investigator. Your answer should address technical competencies, legal/procedural knowledge, and professional conduct.

A digital forensic investigator operates at the intersection of technology, law, and ethics, and effective practice in this field demands a genuinely interdisciplinary skill set that cannot be reduced to technical proficiency alone, however sophisticated.

Technical competencies form the foundation of forensic capability. An investigator must possess deep familiarity with operating system internals — understanding file systems, registry structures (on Windows systems), log file formats, and how data is structured, stored, and sometimes hidden or deleted across different platforms. Equally important is competence with the specific tools that operationalise this knowledge: imaging tools such as **FTK Imager** for creating verified forensic copies, analysis platforms such as **Autopsy** for parsing file systems and recovering deleted data, network analysis tools such as **Wireshark** for examining captured traffic, mobile extraction tools such as **Cellebrite UFED** for recovering data from seized phones, log analysis platforms such as **Splunk** for processing large volumes of server and system logs, and OSINT/link-analysis tools such as **Maltego** for attribution work. Beyond tool operation, genuine technical competence requires the analytical capacity to interpret what these tools reveal — recognising, for instance, that a particular pattern of registry entries indicates a USB device was connected at a specific time, or that a cluster of newly created social media accounts posting near-identical content at coordinated intervals suggests inauthentic, bot-driven amplification rather than organic public reaction.

Legal and procedural knowledge is equally indispensable, since technically excellent forensic work that disregards proper legal procedure can produce findings that are accurate but legally worthless. An investigator must understand the relevant legal framework — in the Cameroonian context, Law No. 2010/012 on Cybersecurity and Cybercriminality — including what specifically constitutes a prosecutable offence, what evidentiary standards courts require, and what procedural safeguards (proper authorisation before accessing private data, adherence to chain-of-custody documentation requirements, correct use of forensically sound collection methods such as write-blocking and hash verification) must be observed at every stage. This legal literacy must extend to practical awareness of jurisdictional complexity, recognising when a case involves cross-border elements that require additional legal mechanisms such as mutual legal assistance requests, and understanding how to structure an investigation's findings into a presentation format (a clear, well-organised report) that a court, including non-technical judges and lawyers, can properly evaluate.

Professional conduct and ethical standards constitute the third essential pillar, and arguably the one most easily overlooked amid technical and legal training, yet equally critical to the integrity of the entire profession. An investigator must maintain rigorous **objectivity**, presenting all relevant findings — including evidence that might complicate or undermine a particular party's preferred narrative — rather than selectively curating results to support a predetermined conclusion; this objectivity is what gives forensic testimony its distinctive credibility and persuasive weight in court, since an investigator perceived as an advocate rather than a neutral technical expert undermines the value of their entire contribution. Investigators must maintain strict **confidentiality**,

given that forensic investigations routinely expose highly sensitive personal, financial, or institutional information that must not be disclosed beyond what the investigation legitimately requires. They must exercise careful **professional judgment regarding the limits of their own expertise and authority**, escalating to specialists or relevant authorities (such as ANTIC in the Cameroonian context, or international partners in cross-border cases) rather than overstepping their competence in pursuit of a result. Finally, investigators must maintain **procedural discipline** even under significant time pressure or institutional urgency — as illustrated by the examination board server breach scenario, where panic and urgency might tempt shortcuts in evidence handling, but where exactly this kind of pressure makes rigorous adherence to proper procedure most important, since rushed or careless work under pressure is precisely when chain-of-custody gaps, improper imaging, or other procedural failures are most likely to occur.

These three pillars — technical competence, legal/procedural knowledge, and ethical professional conduct — are deeply interdependent rather than separable. Technical skill without legal awareness produces evidence that cannot be used; legal awareness without technical skill cannot properly gather or interpret evidence in the first place; and technical and legal competence without ethical discipline can produce biased, selectively presented, or carelessly handled findings that ultimately undermine the credibility of forensic work as a profession. A genuinely effective digital forensic investigator must therefore be simultaneously a skilled technician, a careful student of legal procedure, and a disciplined, trustworthy professional — recognising that excellence in any one dimension cannot compensate for serious deficiency in either of the other two.

Essay 5 (16 marks)

"The greatest challenge in digital forensics is not collecting evidence, but proving that the evidence has not been tampered with." Critically discuss this statement with reference to chain of custody, hashing, and evidence preservation rules.

This statement makes a claim that, upon critical examination, holds substantial merit, though it benefits from some qualification regarding the relative difficulty of evidence collection itself in certain complex cases.

The core insight behind the statement is that **collecting digital evidence, while requiring specialised tools and technical skill, is in many respects the more procedurally straightforward task**, particularly once forensic methodologies and tools (Cellebrite UFED for mobile extraction, FTK Imager for disk imaging, Wireshark for network capture) have matured to the point of being well-documented, widely taught, and reliably executable by trained practitioners following established procedures. The genuinely harder and more legally consequential challenge lies in subsequently *proving*, to a legal standard that can withstand adversarial challenge, that whatever evidence was collected remains an accurate and unaltered representation of its original state throughout the entire period between collection and presentation in court — a period that may span months or years and involve multiple analysts, storage transfers, and analytical procedures, any one of which could introduce a plausible challenge to the evidence's integrity if not meticulously documented and safeguarded.

Chain of custody is the primary mechanism addressing this challenge, providing a continuous, unbroken documentary record of every individual who has had access to a piece of evidence, when, why, and under what conditions, from the moment of initial seizure through to final presentation. The difficulty here is not merely creating this documentation but maintaining its completeness and accuracy under real-world institutional pressures — busy investigators handling multiple cases, evidence storage facilities with imperfect access logging, and the simple human tendency toward administrative shortcuts when workload is high. A single

unexplained gap — a period during which evidence access was not properly logged, or a transfer between custodians that lacks clear documentation — can provide defence counsel with sufficient grounds to argue that tampering *might* have occurred during that window, and many legal systems apply a precautionary standard that favours excluding evidence whenever such a possibility cannot be definitively ruled out, regardless of whether actual tampering can be affirmatively shown to have occurred. This illustrates precisely why the statement's emphasis on *proving* non-tampering, rather than merely *achieving* non-tampering, captures the real difficulty: even evidence that was, in fact, perfectly preserved can still be successfully challenged and excluded if the documentation proving that preservation is itself incomplete or unconvincing.

Hashing provides a more mathematically rigorous complement to chain-of-custody documentation, offering objective, technically verifiable proof — rather than merely procedural assurance — that a piece of evidence has not been altered. By computing a cryptographic hash value at the moment of collection and verifying that this same hash value still matches at every subsequent point the evidence is accessed or presented, investigators can demonstrate non-tampering with a degree of mathematical certainty that purely procedural documentation alone cannot match; even a single altered bit within a multi-gigabyte disk image would produce a completely different hash value, making this an exceptionally sensitive integrity check. However, hashing's power is itself dependent on proper underlying practice: hash values must be computed immediately upon collection (before any possibility of alteration), must be securely recorded in a manner that itself cannot be retroactively altered, and must be consistently re-verified at each subsequent access point — meaning hashing complements rather than replaces the procedural discipline of chain-of-custody documentation.

Evidence preservation rules more broadly — including the use of write-blocking hardware during imaging, secure and access-controlled evidence storage facilities, and formal preservation requests issued promptly to third parties whose data might otherwise be automatically deleted — together form the practical infrastructure that makes both chain-of-custody documentation and hash verification meaningful in the first place. Without proper preservation practices, there would be nothing reliable to document or hash in the first place; preservation, custody, and hashing are therefore best understood not as three independent challenges but as three interlocking components of a single integrated assurance system.

A balanced critical view, then, would suggest that while the statement's central insight is sound — the legal challenge of *proving* integrity is indeed often harder and more consequential than the technical challenge of *achieving* initial collection — this should not be taken to suggest that evidence collection itself is trivial. In genuinely complex cases, such as large-scale database theft spanning multiple servers, or cross-border investigations requiring cooperation from foreign hosting providers, the collection phase itself can be extraordinarily difficult, time-consuming, and legally fraught in its own right. The more precise formulation, therefore, is that digital forensics presents *two* substantial and distinct challenges — technically sound collection, and legally defensible proof of subsequent integrity — and that the latter, while perhaps less visibly "technical," is disproportionately responsible for cases failing at the prosecution stage even when the former has been executed competently, making it an entirely reasonable candidate for the "greatest challenge" the statement identifies.

End of Detailed Essay Answers.